

The Chicxulub Asteroid Impact and Mass Extinction at the Cretaceous-Paleogene Boundary

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The Cretaceous-Paleogene boundary ~65.5 million years ago marks one of the three largest mass extinctions in the past 500 million years. The extinction event coincided with a large asteroid impact at Chicxulub, Mexico, and occurred within the time of Deccan flood basalt volcanism in India. Here, we synthesize records of the global stratigraphy across this boundary to assess the proposed causes of the mass extinction. Notably, a single ejecta-rich deposit compositionally linked to the Chicxulub impact is globally distributed at the Cretaceous-Paleogene boundary. The temporal match between the ejecta layer and the onset of the extinctions and the agreement of ecological patterns in the fossil record with modeled environmental perturbations (for example, darkness and cooling) lead us to conclude that the Chicxulub impact triggered the mass extinction.

Paleontologists have long recognized the global scale and abruptness of the major biotic turnover at the Cretaceous-Paleogene (K-Pg, formerly K-T) boundary ~65.5 million years ago (Ma). This boundary represents one of the most devastating events in the history of life (1) and abruptly ended the age of the dinosaurs. Thirty years ago, the discovery of an anomalously high abundance of iridium and other platinum group elements (PGEs) in the K-Pg boundary clay led to the hypothesis that an asteroid ~10 km in diameter collided with Earth and rendered many environments uninhabitable (2, 3).

The occurrence of an impact is substantiated by the recognition of impact ejecta including spherules, shocked minerals, and Ni-rich spinels in many K-Pg boundary event deposits [e.g., (4, 5)]. The ejecta distribution points to an impact event in the Gulf of Mexico–Caribbean region; this prediction is reinforced by the discovery of the ~180- to 200-km-diameter Chicxulub crater structure on the Yucatan peninsula, Mexico (6). Modeling suggests that the size of the crater and the release of climatically sensitive gases from the carbonate- and sulfate-rich target rocks could have caused catastrophic environmental effects

such as extended darkness, global cooling, and acid rain (7–9). These effects provide an array of potential mechanisms for the ecologically diverse but selective abrupt extinctions (Fig. 1) (10–13).

Notwithstanding the substantial evidence supporting an impact mechanism, other interpretations of the K-Pg boundary mass extinction remain. Stratigraphic and micropaleontological data from the Gulf of Mexico and the Chicxulub crater have instead been used to argue that this impact preceded the K-Pg boundary by several hundred thousand years and therefore could not have caused the mass extinction [e.g., (14)]. In addition, the approximately one-million-year-long emplacement of the large Deccan flood basalts in India spans the K-Pg boundary (Fig. 1); the release of sulfur and carbon dioxide during these voluminous eruptions may have caused severe environmental effects (15) that have also been proposed as triggers for the mass extinction at the K-Pg boundary (16).

Here, we assess the observational support for these divergent interpretations by synthesizing recent stratigraphic, micropaleontological, petrological, and geochemical data from the globally distributed K-Pg boundary event deposit. Impact and volcanism as extinction mechanisms are evaluated in terms of their predicted environmental perturbations and, ultimately, the distribution of life on Earth before and after the K-Pg boundary.

What Is the Evidence for Correlating the Impact with the K-Pg Boundary?

The Upper Cretaceous and lower Paleogene sediments bracketing the K-Pg boundary event deposits are among the most intensively investigated deposits in the geological record. More than 350 K-Pg boundary sites are currently known, and these sites show a distinct ejecta distribution pattern related to distance from the Chicxulub crater (Fig. 2 and table

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